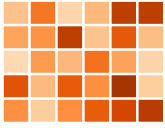
The claim

Brain map y
(one value per parcel)



Gene set S: expression of its genes across the parcels

T = correlation of the map with the set's mean expression

Two nulls, three checks

Spatial leg
Is T larger than for smooth surrogate maps (class-stratified)?

Gene / species leg
Is T larger than for gene sets as coherent and as brain-expressed as S? (four nulls; weakest result counts)

Level 0
Is T nonzero against the naive reference?

Fourth-corner max test

Report an association only if all three checks pass, each at alpha = .025: level 0 AND spatial leg AND weakest gene/species result

Verdict: survives, or not detectable at this power

Why two legs: a map can match a gene set through shared spatial smoothness (spatial leg) or through gene coherence and brain-wide expression (gene/species leg). Each null alone is blind to the other route.

Schematic; no data. The coloured grids are decorative, not results.